
Classical Forecasting vs. Time-Series Foundation Models

Comparing established methods and pretrained models across 12 datasets

5 classical methods

3 foundation-model families

12 datasets

Prepared by

Amirhossein Bagheri

Politecnico di Milano

Supervisor: Tommaso Bianchi

Why this comparison matters

Do time-series foundation models **consistently** outperform classical forecasting approaches across datasets with different statistical characteristics?

Zero-shot promise

Can a pretrained model transfer across domains, frequencies, and horizons without dataset-specific training?

Model families

Compare that promise against established statistical, algorithmic, and neural forecasting approaches.

Different metrics

Separate point accuracy from quantile and predictive-distribution quality; they answer different questions.

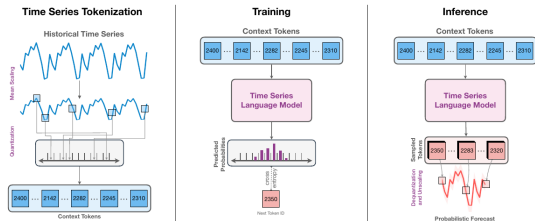
Statistical tests

Test whether observed performance differences are systematic or compatible with sampling variation.

Classical and conventional forecasting models

Model	Forecasting mechanism	Main strength	Main limitation
Seasonal Naive	Repeats the most recent seasonal cycle.	Transparent and robust reference for seasonal data.	Cannot adapt to trend or changing seasonal structure.
ARIMA (2, 1, 2)	Models differenced values through autoregressive and moving-average terms.	Captures short-range linear dynamics with a compact local model.	Fixed order; limited under nonlinear behavior or regime changes.
AutoARIMA	Searches candidate ARIMA orders and selects the best fit by AIC.	Automates model specification for each series.	Search can be costly or unstable on short and noisy histories.
Prophet	Decomposes a series into trend, changepoints, and calendar seasonalities.	Interpretable components and flexible trend changes.	A rigid additive decomposition may miss local dynamics.
DeepAR	Learns an autoregressive recurrent network under a probabilistic likelihood.	Represents nonlinear temporal relationships and uncertainty.	Requires training and hyperparameter choices for each task.

Chronos: forecasting as a language-model task



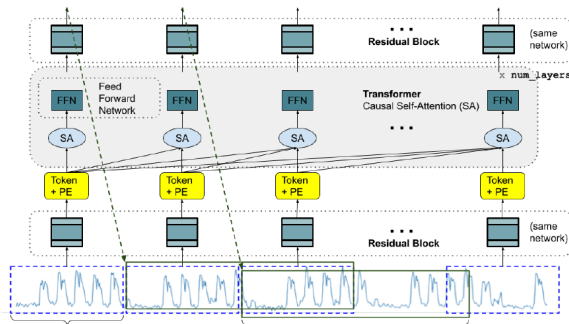
- **Representation:** mean-scale the series, quantize values into bins, and map them to a fixed token vocabulary.
- **Architecture:** a T5 encoder–decoder learns next-token probabilities using cross-entropy.
- **Forecast:** autoregressive token samples are dequantized into trajectories. Inference is **stochastic**; the sampled trajectories define a **probabilistic forecast**.

Chronos–T5 variants

Variant	Params	Enc./Dec.
Tiny	8M	4 / 4
Mini	20M	4 / 4
Small	46M	6 / 6
Base	200M	12 / 12
Large	710M	24 / 24

The variants use the same tokenization and forecasting mechanism. Capacity increases through greater depth and width.

TimesFM: patch the signal, decode the future



Patch tokens

Non-overlapping blocks of 32 observations are normalized and projected into token embeddings.

Decoder-only

Causal self-attention lets each patch representation use only its available history.

Multi-horizon

Each output head predicts a block of future time points; blocks are composed to reach the requested horizon.

Forecast behavior

TimesFM 2.5 directly outputs the mean and quantiles q10–q90. The values are **probabilistic forecasts**, but inference is **deterministic**: the same input returns the same distribution.

Moirai: one model for any frequency and variate count



Multiple patch sizes

A bank of input/output projections supports several patch lengths. The model can match its temporal resolution to series with different sampling frequencies and local dynamics.

Any-variate attention

Patches from every variable are flattened into one token sequence. Time and variable identifiers preserve where each patch came from, so attention can learn both within- and cross-variable dependence.

Distribution forecast

Masked future patches are decoded into an uncertainty representation. The original Moirai uses a flexible mixture density; Moirai 2.0 directly predicts quantiles. Inference is deterministic.

Twelve datasets I: competition and archive panels

Dataset	Domain	Full dataset	Series used	Freq. / H	Observations	Statistical characteristics
M1 Yearly	Business / economic	181 univariate series	20	Y / 6	15–58	Trends, level shifts, weak or absent seasonality
M3 Quarterly	Business / economic	756 univariate series	20	Q / 8	24–72 points	Quarterly cycles, trend, nonstationarity
M4 Hourly	Mixed competition	414 univariate series	20	h / 48	748–1,008 points	High frequency, daily seasonality, noise
Weather (rain)	Environment	Rain subset of 3,010 series	20	D / 30	1,332–65,981	Zero inflation, intermittency, seasonal weather
Car Parts	Retail / inventory	2,674 univariate series	20	M / 12	51 each	Sparse intermittent demand, many zeros
NN5 Weekly	Banking / ATM	111 univariate series	20	W / 8	about 113	Annual seasonality, holidays, demand volatility

Twelve datasets II: health, infrastructure, and macroeconomics

Dataset	Domain	Full dataset	Series used	Freq. / H	Observations	Statistical characteristics
National Illness	Public health	1 series, 7 variables	1 target	W / 24	966 weeks	Epidemic peaks, annual cycle, regime change
Traffic	Transport	1 series, 862 sensors	5 targets	h / 24	17,544 hours	Strong daily/weekly cycles, correlated sensors
PSM	Server telemetry	1 series, 25 variables	5 targets	h / 24	132,481 min. rows, resampled h	Anomalies, abrupt shifts, high-frequency noise
Elexon Demand	Energy	1 national demand series	1	D / 30	2024–2026 (730 days)	Weekly cycle, weather/calendar effects, nonstationarity
USGS Streamflow	Hydrology	5 gauge series	5	D / 30	2015–2026 (~4,230 days)	Annual hydrology, skew, floods/droughts, gaps
BLS Macro	Macro	3 indicator series	3	M / 6	2010–2026 (~199 months)	Trends, structural breaks, heterogeneous scales

Point and probabilistic forecast evaluation

MASE point

$$\text{MASE} = \frac{H^{-1} \sum_{h=1}^H |y_h - \hat{y}_h|}{(T-m)^{-1} \sum_{t=m+1}^T |y_t - y_{t-m}|}$$

Error scaled by an in-sample seasonal-naïve error. It is comparable across units and datasets.

< 1 beats the scale; $= 1$ matches it; > 1 is worse.

Central forecast

How accurate is the median path?

WQL quantiles

$$\text{WQL} = \frac{1}{|Q|} \sum_{q \in Q} \frac{2 \sum_h \rho_q(y_h - \hat{y}_{q,h})}{\sum_h |y_h|}$$

Weighted quantile (pinball) loss on $Q = \{0.1, \dots, 0.9\}$, normalized by target magnitude.

Multiple quantiles reveal asymmetric risk and interval quality.

Quantile accuracy

Where are under- and over-predictions?

CRPS distribution

$$\text{CRPS}(F, y) = \int_{-\infty}^{\infty} (F(z) - \mathbf{1}\{y \leq z\})^2 dz$$

Scores the entire predictive CDF against the observation; rewards **calibration** and **sharpness** jointly.

Implemented as a finite-quantile approximation from q10–q90.

Distribution quality

How well does the predictive CDF match reality?

Together they separate scale-free point error from quantile and distributional quality; lower is better.

Statistical inference for forecast comparisons

Diebold–Mariano

Which of two models is better?

$$d_t = L_t(A) - L_t(B), \quad H_0 : \mathbb{E}[d_t] = 0$$

Use: compare A and B on the same forecast origins. The variance estimate accounts for serial dependence in d_t .

Decision

$p < \alpha$ and $\bar{d} < 0$: A has significantly lower expected loss.

SPA

Does any alternative beat a benchmark?

$$d_{k,t} = L_t(B) - L_t(k)$$

$$H_0 : \max_k \mathbb{E}[d_{k,t}] \leq 0$$

Use: compare benchmark B with many challengers at once. The bootstrap corrects for searching over multiple models.

Decision

Reject H_0 : at least one challenger has lower expected loss than B.

Model Confidence Set

Which models remain plausible best models?

$$\mathcal{M}_0 \supset \cdots \supset \widehat{\mathcal{M}}_{1-\alpha}$$

Use: begin with every model. Repeatedly test equal predictive ability and remove the worst-supported model.

Output

$\widehat{\mathcal{M}}_{1-\alpha}$ is a set of models that cannot be distinguished from the best at level α .

Empirical winners across the three core losses

Dataset	MASE	WQL	CRPS
M1 Yearly	TimesFM	ARIMA	ARIMA
M4 Hourly	Chronos Base	Chronos Base	Chronos Base
Weather	Chronos Base	Moirai	Moirai
M3 Quarterly	TimesFM	TimesFM	TimesFM
Car Parts	Moirai	Moirai	Moirai
NN5 Weekly	TimesFM	TimesFM	TimesFM
National Illness	TimesFM	TimesFM	TimesFM
Traffic	Moirai	Moirai	Moirai
PSM	Chronos Small	Chronos Small	Chronos Small
Elexon Demand	TimesFM	TimesFM	TimesFM
USGS Streamflow	Moirai	TimesFM	TimesFM
BLS Macro	Auto-ARIMA	TimesFM	TimesFM

foundation model

statistical model

Each cell is the lowest aggregate loss among 11 models; this is a **descriptive ranking**, not a significance claim.

How often each model records the lowest loss

Model	MASE	WQL	CRPS	Total
TimesFM 2.5 200M	5	6	6	17
Moirai 2.0 Small	3	3	3	9
Chronos Base	2	1	1	4
Chronos Small	1	1	1	3
ARIMA(2,1,2)	0	1	1	2
Auto-ARIMA	1	0	0	1
Chronos Mini	0	0	0	0
Chronos Tiny	0	0	0	0
DeepAR	0	0	0	0
Prophet	0	0	0	0
Seasonal Naive	0	0	0	0
Column total	12	12	12	36

36 comparisons

- ▶ Each model is ranked on **MASE, WQL, and CRPS** across 12 datasets.
- ▶ A win means recording the lowest aggregate loss for one dataset under one metric.
- ▶ TimesFM records the lowest loss **17 times**; Moirai does so **9 times**.
- ▶ Across all comparisons, a foundation model ranks first **33 times**.

When every foundation model beats every comparator

Strict family dominance: $\max_{m \in \text{FM}} L(m) < \min_{m \in \text{C}} L(m)$ — even the weakest foundation model has lower aggregate loss than the strongest classical/conventional comparator.

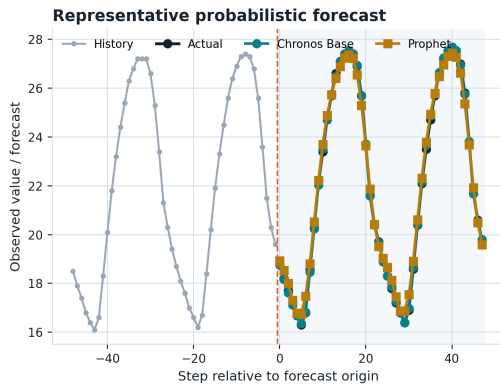
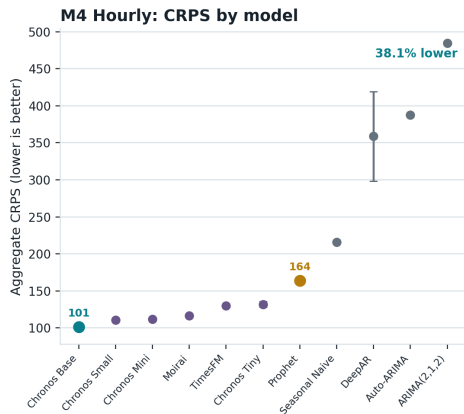
Dataset	MASE	WQL	CRPS
M4 Hourly	✓ Chronos Base	✓ Chronos Base	✓ Chronos Base
PSM [†]	✓ Chronos Small	✓ Chronos Small	✓ Chronos Small
Elexon Demand [†]	✓ TimesFM	✓ TimesFM	✓ TimesFM
USGS Streamflow [†]	✓ Moirai	✓ TimesFM	✓ TimesFM
Car Parts	<i>not strict</i>	✓ Moirai	✓ Moirai
Traffic [†]	<i>not strict</i>	✓ Moirai	✓ Moirai

16 strict comparisons

Four datasets satisfy the condition for all three losses; Car Parts and Traffic satisfy it for WQL and CRPS.

[†] Descriptive low-power case; among the four complete-dominance datasets, only M4 Hourly has at least 20 paired tasks.

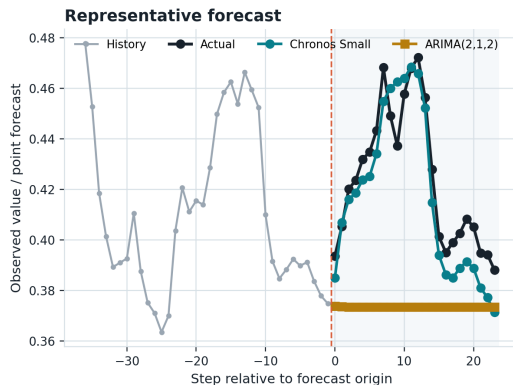
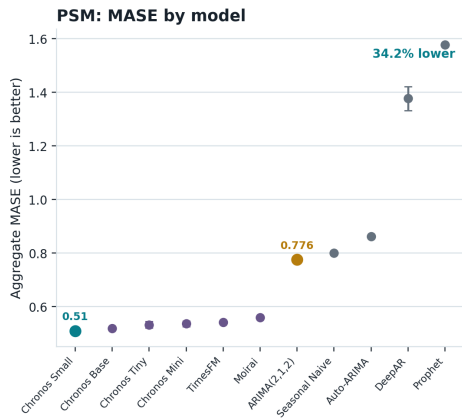
Clear empirical separation: M4 Hourly CRPS



38.1% lower aggregate CRPS

Similar point forecasts can still have differently scored distributions.

Clear empirical separation: PSM MASE



34.2% lower aggregate MASE

Chronos Small follows the rise and decline; ARIMA remains nearly flat.

Where a foundation model ranks first in a mixed ordering

Mixed ordering: a foundation model has the lowest loss, but at least one classical/conventional model still outranks another foundation model.

Dataset	MASE	WQL	CRPS
Car Parts	Moirai	<i>strict FM</i>	<i>strict FM</i>
M1 Yearly	TimesFM	<i>classical first</i>	<i>classical first</i>
M3 Quarterly	TimesFM	TimesFM	TimesFM
NN5 Weekly	TimesFM	TimesFM	TimesFM
Weather	Chronos Base	Moirai	Moirai
National Illness [†]	TimesFM	TimesFM	TimesFM
Traffic [†]	Moirai	<i>strict FM</i>	<i>strict FM</i>
BLS Macro [†]	<i>classical first</i>	TimesFM	TimesFM

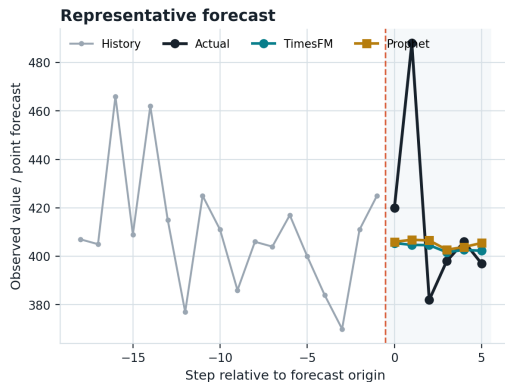
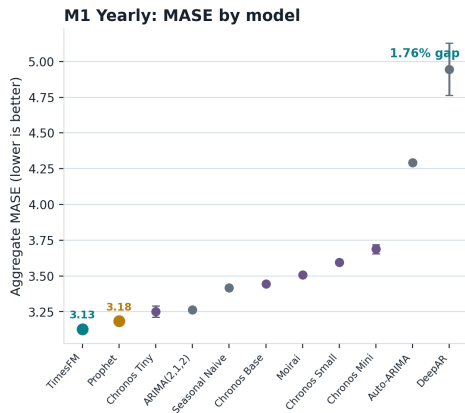
17 mixed comparisons

strict FM dominance

classical model ranks first

Purple cells name the empirical foundation-model leader. [†] Descriptive low-power dataset.

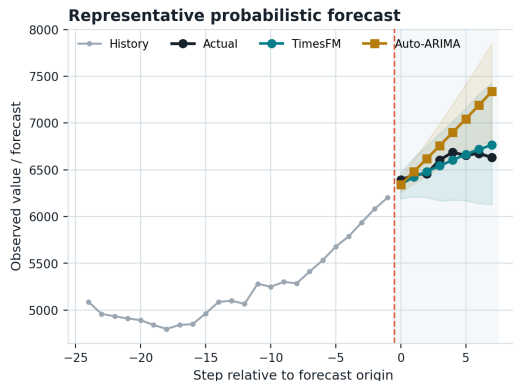
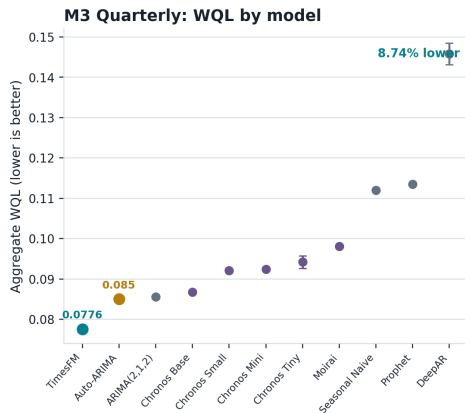
Close empirical contest: M1 Yearly MASE



1.76% aggregate gap

TimesFM ranks first, while Prophet is close and the forecasts nearly overlap.

Close empirical contest: M3 Quarterly WQL



8.74% lower aggregate WQL

TimesFM follows the realized level; Auto ARIMA drifts upward.

Where a classical model beats every foundation model

3 of 36 cells

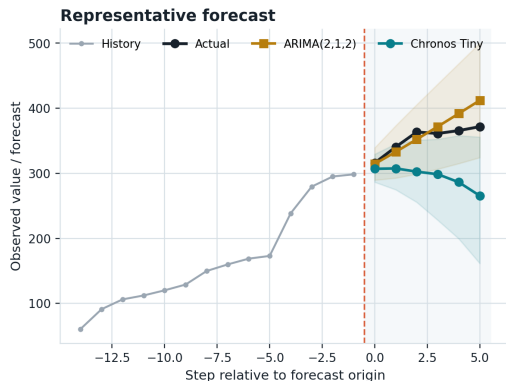
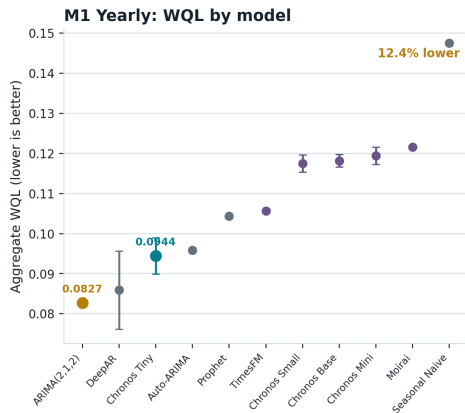
Dataset	Loss	Classical winner		Best foundation model		Lower loss
M1 Yearly	WQL	ARIMA	0.0827	Chronos Tiny	0.0944	12.4%
M1 Yearly	CRPS	ARIMA	202,882	Chronos Tiny	231,712	12.4%
BLS Macro [†]	MASE	Auto-ARIMA	0.141	TimesFM	0.174	18.8%

M1 is the substantive exception
ARIMA leads both probabilistic losses
across 60 paired tasks.

BLS remains tentative
Its MASE result is based on only seven
paired tasks.

Values are aggregate losses; percentages compare the classical winner with the best-performing foundation model. Lower is better.

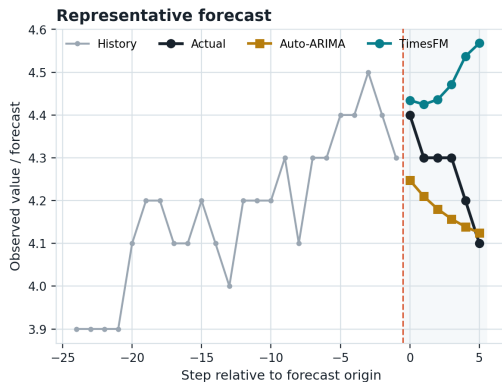
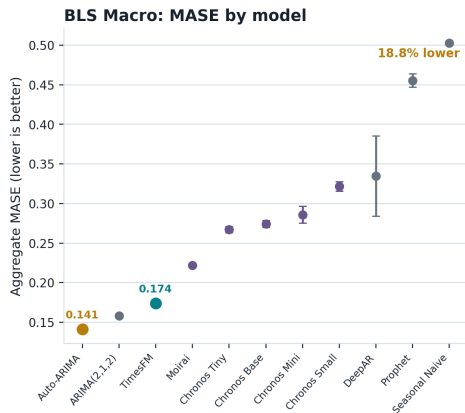
Classical exception: M1 Yearly WQL



12.4% lower aggregate WQL

ARIMA follows the rise; its 10–90% interval covers the observations.

Classical exception: BLS Macro MASE



18.8% lower aggregate MASE

Auto-ARIMA follows the decline; TimesFM extrapolates upward.

How to read the repetition results

p_{Holm} is a **corrected p-value**. Holm correction accounts for testing several dataset and metric combinations at the same time.

$$p_{\text{Holm}} < 0.05$$

The observed lead is large relative to seed variation in the three runs.

SI

The same positive gap occurs in every run; a finite p-value is undefined.

$$p_{\text{Holm}} \geq 0.05$$

Three repetitions do not distinguish the lead from seed variation.

Example	Reported value	Interpretation
M4 Hourly CRPS	$p = 0.010$	The lead remains clear after correcting for multiple tests.
M1 Yearly WQL	$p = 0.068$	The observed lead is not resolved by three repetitions.
BLS Macro MASE	SI	The same positive gap is recorded in every repetition.

These results describe repeatability across the three random seeds.

Do the rankings persist across random seeds?

Winner versus the best model from the opposite family: $d_r = L_{\text{opposite},r} - L_{\text{winner},r}$, $H_0 : \mathbb{E}[d] \leq 0$, one-sided paired t test, $n = 3$.

Dataset	MASE	WQL	CRPS
Car Parts	0.025	SI	SI
M1 Yearly	0.004	0.068	0.068
M3 Quarterly	SI	SI	SI
M4 Hourly	< 0.001	0.010	0.010
NN5 Weekly	0.047	0.025	0.025
Weather	0.091	0.025	0.025
National Illness	< 0.001	< 0.001	< 0.001
PSM	< 0.001	< 0.001	< 0.001
Traffic	SI	SI	SI
BLS Macro	SI	SI	SI
Elexon Demand	SI	SI	< 0.001
USGS Streamflow	SI	0.052	0.052

17 significant

14 seed invariant

5 unresolved

Holm-adjusted p values cover the 22 finite tests. SI means the same positive gap occurred in all three runs, so finite t and p values do not exist. The table measures repeatability across the three random seeds only.

Is the empirical benchmark narrative enough?

Are the observed winners **meaningfully** winning?

Empirical narrative

- ▶ Lowest aggregate metric values
- ▶ Average ranks and winner counts
- ▶ Breadth across datasets and domains
- ▶ Point, quantile, and distributional losses

Unanswered questions

- ▶ Is the observed gap distinguishable from sampling variation?
- ▶ Does the claim survive comparison with ten alternatives?
- ▶ Which models remain statistically plausible winners?
- ▶ Are small-sample results sufficiently powered?

empirical winner



SPA + DM



95% MCS



defensible claim

MASE: statistical evidence behind the empirical winner

Dataset	Empirical winner	SPA	DM*	MCS size
M1 Yearly	TimesFM	3/10	1/10	9
M4 Hourly	Chronos Base	5/10	10/10	6
Weather	Chronos Base	4/10	6/10	8
M3 Quarterly	TimesFM	9/10	4/10	4
Car Parts	Moirai	4/10	4/10	7
NN5 Weekly	TimesFM	10/10	10/10	1
National Illness [†]	TimesFM	1/10	—	7
Traffic [†]	Moirai	5/10	3/10	6
PSM [†]	Chronos Small	4/10	4/10	7
Elxon Demand [†]	TimesFM	9/10	—	3
USGS Streamflow [†]	Moirai	3/10	1/10	8
BLS Macro [†]	Auto-ARIMA	5/10	—	5

SPA/DM = Holm-adjusted significant wins out of 10; MCS = models retained in the 95% set. *DM is exploratory; — = not run (< 8 ordered tasks). [†]Low power (< 20 paired tasks).

Strict separation occurs only on NN5 Weekly: TimesFM beats all ten alternatives by SPA and is the sole MCS member.

WQL: statistical evidence behind the empirical winner

Dataset	Empirical winner	SPA	DM*	MCS size
M1 Yearly	ARIMA	0/10	0/10	11
M4 Hourly	Chronos Base	0/10	0/10	11
Weather	Moirai	9/10	9/10	2
M3 Quarterly	TimesFM	9/10	7/10	2
Car Parts	Moirai	4/10	5/10	7
NN5 Weekly	TimesFM	10/10	10/10	1
National Illness [†]	TimesFM	1/10	—	5
Traffic [†]	Moirai	6/10	3/10	5
PSM [†]	Chronos Small	4/10	3/10	7
Elxon Demand [†]	TimesFM	9/10	—	2
USGS Streamflow [†]	TimesFM	0/10	0/10	11
BLS Macro [†]	TimesFM	0/10	—	11

SPA/DM = Holm-adjusted significant wins out of 10; MCS = models retained in the 95% set. *DM is exploratory; — = not run (< 8 ordered tasks). [†]Low power (< 20 paired tasks).

Strict separation occurs only on NN5 Weekly; Weather and M3 show strong but not unique leadership.

CRPS: statistical evidence behind the empirical winner

Dataset	Empirical winner	SPA	DM*	MCS size
M1 Yearly	ARIMA	0/10	0/10	11
M4 Hourly	Chronos Base	0/10	0/10	11
Weather	Moirai	10/10	10/10	1
M3 Quarterly	TimesFM	9/10	7/10	5
Car Parts	Moirai	4/10	10/10	7
NN5 Weekly	TimesFM	10/10	10/10	1
National Illness [†]	TimesFM	1/10	—	5
Traffic [†]	Moirai	6/10	3/10	5
PSM [†]	Chronos Small	4/10	3/10	7
Elxon Demand [†]	TimesFM	9/10	—	2
USGS Streamflow [†]	TimesFM	0/10	0/10	11
BLS Macro [†]	TimesFM	1/10	—	11

SPA/DM = Holm-adjusted significant wins out of 10; MCS = models retained in the 95% set. *DM is exploratory; — = not run (< 8 ordered tasks). [†]Low power (< 20 paired tasks).

Weather (Moirai) and NN5 Weekly (TimesFM) are the two clean CRPS results: 10/10 SPA wins and singleton confidence sets.

Conclusion: what the empirical results show

33 of 36

foundation model ranks first

16

strict foundation dominance

3

classical model ranks first

Foundation models usually achieve the lowest aggregate loss, **but the pattern is not uniform.**

The general pattern

- ▶ TimesFM records the most empirical wins.
- ▶ Chronos and Moirai lead on several datasets where the separation from classical methods is substantial.

The important exceptions

- ▶ ARIMA has the lowest WQL and CRPS on M1 Yearly.
- ▶ Auto ARIMA has the lowest MASE on BLS Macro.
- ▶ Some foundation model wins are relatively close, as seen for M1 MASE and M3 WQL.

Conclusion: what the statistical analysis adds

4 comparisons show complete statistical separation

NN5 MASE, NN5 WQL, NN5 CRPS, and Weather CRPS

What is well supported

- ▶ NN5 provides the clearest evidence. TimesFM separates from every alternative for all three losses.
- ▶ Moirai also separates from every alternative for Weather CRPS.
- ▶ These conclusions agree across adjusted SPA tests and the model confidence set.

What remains uncertain

- ▶ Most empirical winners remain statistically tied with at least one alternative.
- ▶ Several datasets contain too few paired tasks for a strong claim.
- ▶ A broad win count therefore says more about coverage than about certainty.

Overall: Foundation models are strong defaults when no retraining is available. Classical models still matter, and each claimed improvement should be checked for the dataset, the loss, and the available evidence.